OTH REGENSBURG

Praktikum Regelungstechnik (RTP)

Versuch T5:
RetBot, distance control

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Gruppe 38
Marius Off
Janis Graf

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1 Preparations

1.1. (aV) P controller in the time domain

(aV) What is the equation that describes a digital P controller in the time domain? [1]

$$x_a[t] = K_P \cdot e[t] \tag{1.1}$$

1. $x_a(t)$: Input variable of the P controller
2. K_p: Proportional coefficient
3. $y(t)$: Control output of the P controller

1.2. (bV) Transfer function of open loop P controller

(bV) What is the transfer function of the open loop with P controller?
Which characteristic shows $G_o(s)$? [1]

The transfer function $G_o(s)$ is obtained using the block diagram determined in the fourth experiment as follows:

$$G_o(s) = K_P \cdot \frac{-K_I}{T_1 \cdot s^2 + s} \tag{1.2}$$

$G_o(s)$ exhibits IT1 behavior.

1.3. (cV) Transfer function of distance control with P controller

(cV) What is the transfer function of the distance control with P controller?
Which characteristic shows $G_W(s)$? [1]

By inserting 1.2, we obtain the following for $G_W(s)$:

$$G_W(s) = \frac{G_o(s)}{1 + G_o(s)} = \frac{1}{1 + \frac{-T_1}{K_P \cdot K_I} \cdot s^2 - \frac{1}{K_P \cdot K_I} \cdot s} \tag{1.3}$$

The closed-loop system $G_W(s)$ exhibits PT_2 behavior.

1.4. (dV) Reasonable damping ratio D for the closed control loop

(dV) Select a reasonable damping ratio D for the closed control loop and calculate the controller parameter K_P so that the desired/selected damping ratio is achieved. [1]

The damping ratio is set to D = 1 (aperiodic limit case) to ensure a fast, overshoot-free settling. This allows the robot to reach the target distance as quickly as possible without having to move backward. The standard mathematical form of a PT2 system is:

$$G_s = \frac{K_P}{\frac{1}{\omega_0^2} \cdot s^2 + \frac{2D}{\omega_0 \cdot s} + 1} \tag{1.4}$$

Rearranging 1.4 yields:

$$\frac{1}{\omega_0^2} = -\frac{T_1}{K_P \cdot K_I} \rightarrow \omega_0 = \sqrt{-\frac{K_P \cdot K_I}{T_1}} \tag{1.5}$$

$$\frac{2D}{\omega_0} = -\frac{1}{K_P \cdot K_I} \rightarrow \omega_0 = -K_P \cdot K_I \cdot 2 \cdot D \tag{1.6}$$

If we now set Equation 1.4 equal to 1.4, we can solve for K_P.

$$\sqrt{-\frac{K_P \cdot K_I}{T_1}} = -K_P \cdot K_I \cdot 2 \cdot D \rightarrow K_P = \frac{1}{4 \cdot D^2 \cdot T_1 \cdot K_I} \tag{1.7}$$

Using the previously determined parameters (mean value) K_I and T_1,
Left motor:

$$K_{I,L} = 0{,}437\,\mathrm{cm\,s^{-1}\,\%^{-1}}, \qquad T_{1,L} = 57{,}1\,\mathrm{ms} = 0{,}0571\,\mathrm{s} \tag{1.8}$$

Right motor:

$$K_{I,R} = 0{,}446\,\mathrm{cm\,s^{-1}\,\%^{-1}}, \qquad T_{1,R} = 52{,}2\,\mathrm{ms} = 0{,}0522\,\mathrm{s} \tag{1.9}$$

the following values are obtained:
Left motor:

$$K_{P,L} = \frac{1}{4 \cdot 1^2 \cdot 0.0571 \cdot 0.437} = 9{,}99\,\%\,\mathrm{cm^{-1}} \approx 10\,\%\,\mathrm{cm^{-1}} \tag{1.10}$$

Right motor:

$$K_{P,R} = \frac{1}{4 \cdot 1^2 \cdot 0.0522 \cdot 0.446} = 10{,}77\,\%\,\mathrm{cm^{-1}} \approx 11\,\%\,\mathrm{cm^{-1}} \tag{1.11}$$

2 Experiment T5: RetBot, distance control

2.1. e)

(e) Simulate the idealized behavior of the closed control loop in Matlab/Simulink using the controller parameter K_P calculated in the preparation by using the setup from previous experiment. [1]

2.2. f)

(f) Plot the reference variable, the manipulated variable, and the controlled variable via Matlab (clean graphs, no screenshots from the scope). Is the damping level you selected correct? Interpret the signal curves. [1]



Plot of the reference variable, the manipulated variable, and the controlled variable



influence of the dampening level plot

2.3. g)

(g) Implement the P controller for the LEGO robot - see 3.7 for the existing software-based control loop, that can be parameterized by using following commands using Serial.ICE:

- *'1': set K_P*
- *'2': set K_I*
- *'3': set K_D*
- *'4': set desired distance/cm*
- *'5': set K_0*
- *'6': set K_1*

Show the correct implementation by Serial.ICE commands and received data. [1]

2.4. h)

(h) Upload a video showing the robot moving towards an obstacle. [1]



Link to the video

2.5. i)

(i) Read the recorded data of your test-drive using Serial.ICE and compare it to the simulated results - distance, travelled distance and duty-cycle of motor 0! This has to be done in one graph after importing the data according to the previous experiment! [1]



Comparison of the recorded data of the test-drive and the simulated results - distance, travelled distance and duty-cycle of motor 0

2.6. j)

(j) Increase K_P, check the influence of the rate-limit-factor in a way that the robot shows periodic oscillations - write down this critical value(s) and upload a video. [1]

A Appendix Frontmatter

A.1. Declarations

1. Wir erkläre hiermit, dass wir dieses Dokument selbstständig verfasst, noch nicht anderweitig für Prüfungszwecke vorgelegt, keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie wörtliche und sinngemäße Zitate als solche gekennzeichnet haben.

Regensburg, den 2026-05-01,

”	Marius Off	Matrikelnummer: 3416002,
”	Janis Graf	Matrikelnummer: 3377251

Ort, Datum und Unterschrift

Abbildungsverzeichnis

B Appendix Code and Resources

B.1. Code used for the analysis of the Preperations

Editorial note — Source code omitted

The full source code for the analysis (a Jupyter notebook of this time approximately 1,500 lines and a simulink model) is ***not*** included in this appendix. Including it as a verbatim listing caused major build instability again and was therefore deliberately omitted.
The complete source can be obtained by either of the following means:

- By email request to `janis.graf@st.oth-regensburg.de`
- Via requesting access to the GitHub repository: `https://github.com/janisgra/RTP-T4-SoSe2026.git`

Literatur

[1] Torsten Reitmeier. *Internship in Control Engineering (RTP)*. Praktikumsversuche. 2025.